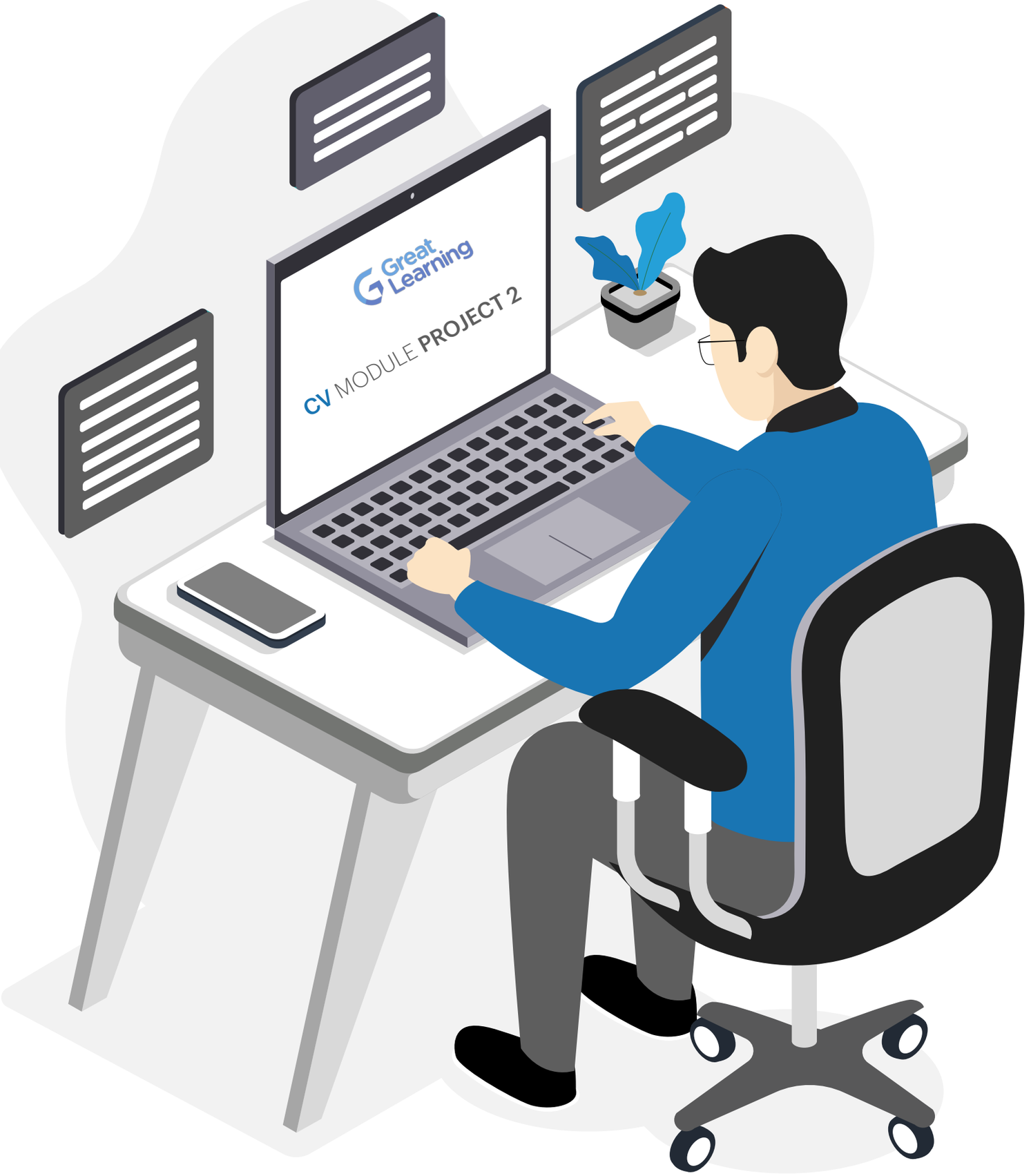


AIML | MODULE PROJECT

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Computer Vision

TOTAL SCORE 30

General Instructions:

1. Submission of all the parts is expected in 1 notebook only
2. Expected submission format: 1 '.ipynb' notebook and 1 '.html' notebook only
3. 50% marks will be deducted if insights/steps are missing in the corresponding questions.
4. If output for any code cell is missing, 50% marks will be deducted.
5. Any kind of Plagiarism will lead to 0 (zero) Marks.

Submission Format:

1. '.ipynb' (Jupyter Notebook) and
2. '.html' (Jupyter Notebook > File > Download as > HTML)
5 Marks will be deducted if submission in any of the formats is missing.

- **DOMAIN:** Botanical Research
- **CONTEXT:** University X is currently undergoing some research involving understanding the characteristics of flowers. They already have have invested on curating sample images. They require an automation which can create a classifier capable of determining a flower's species from a photo.
- **DATA DESCRIPTION:** The dataset comprises of images from 17 plant species.
- **PROJECT OBJECTIVE:** To experiment with various approaches to train an image classifier to predict type of flower from the image.

Steps and tasks: [Total Score: 30 Marks]

1. Import and Understand the data [5 Marks]
   A. Import and read oxflower17 dataset from tflearn and split into X and Y while loading. [2 Marks]
      Hint: It can be imported from tflearn.datasets. If tflearn is not installed, install it.
      It can be loaded using: x, y = oxflower17.load_data()
   B. Print Number of images and shape of the images. [1 Marks]
   C. Print count of each class from y. [2 Marks]

2. Image Exploration & Transformation [Learning purpose - Not related to final model] [10 Marks]
   A. Display 5 random images. [1 Marks]
   B. Select any image from the dataset and assign it to a variable. [1 Marks]
   C. Transform the image into grayscale format and display the same. [3 Marks]
   D. Apply a filter to sharpen the image and display the image before and after sharpening. [2 Marks]
   E. Apply a filter to blur the image and display the image before and after blur. [2 Marks]
   F. Display all the 4 images from above questions besides each other to observe the difference. [1 Marks]

3. Model training and Tuning: [15 Marks]
   A. Split the data into train and test with 80:20 proportion. [2 Marks]
   B. Train a model using any Supervised Learning algorithm and share performance metrics on test data. [3 Marks]
   C. Train a model using Neural Network and share performance metrics on test data. [4 Marks]
   D. Train a model using a basic CNN and share performance metrics on test data. [4 Marks]
   E. Predict the class/label of image 'Prediction.jpg' using best performing model and share predicted label. [2 Marks]

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